

YueD (Bacillus cereus, UniProt Q8RJB2) — Native Substrate & Pocket Hypothesis

Focus: `native-substrate-pocket-alphaFold` (free-text) **Seed hypothesis:** The physiological substrate of YueD is not benzil but a naturally occurring aromatic carbonyl compound, *most plausibly a quinone*, and the AlphaFold substrate pocket should *discriminate between candidate substrate classes*.

Note on this version: Iteration 3 retrieved the actual Km/kcat constants from UniProt's BIOPHYSICOCHEMICAL PROPERTIES block (ECO:0000269,).

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These data **correct** an iteration-2 misreading in which the UniProt FUNCTION-field phrase "in decreasing order" was taken as a substrate-preference ranking. The quantitative kinetics actually **support** the seed's premise that 1,4-naphthoquinone has a far lower Km than benzil. The judgment below reflects the corrected analysis.

Executive Judgment

Verdict: Partially supported. The seed's structural/kinetic reasoning holds up better than a first pass suggested, but the specific identification of the native substrate as "a quinone" remains unresolved.

- **Supported — "not benzil":** Benzil is a wholly synthetic screening substrate

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) and the weakest-binding diketone tested (Km = 768 μM). GO's obsolescence of the benzil-specific term is justified.

- **Supported — pocket/kinetics alignment:** The measured Km(1,4-naphthoquinone) = 27.6 μM is ~28× lower than Km(benzil) = 768 μM

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The AlphaFold pocket is enclosed and aromatic/hydrophobic, sterically disfavoured by the bulky, twisted benzil (10 Å, plane-RMSD 0.39 Å) relative to a flat, compact aromatic such as naphthoquinone. **The pocket geometry I measured is consistent with the stated Km ordering** — i.e., the pocket does appear to discriminate on steric grounds, exactly as the seed proposed.

- **Unresolved — "most plausibly a quinone":** By catalytic efficiency (kcat/Km), the single best in-vitro substrate is **1-phenyl-1,2-propanedione (3.93 min⁻¹μM⁻¹)**, ~15× **better than 1,4-naphthoquinone (0.268)** and ~47× **better than benzil (0.084)** — and it is also a synthetic aryl-alkyl diketone, not a quinone. Methylglyoxal and sepiapterin were never assayed. So the kinetics favour **small/flat aromatic carbonyls in general**, consistent with (but not uniquely pointing to) a quinone.
- **Competing signal — structural homology:** Foldseek's nearest characterized neighbours are **sepiapterin reductases and SDR ketoreductases, not quinone reductases**. So evolutionary context points to pterin/keto-reductase chemistry rather than a dedicated quinone reductase.
- **Mechanistic caveat:** In the homologous human SPR, quinone turnover can be NADPH-dependent **redox cycling at the cofactor site** rather than substrate-pocket reduction (P 23640889).
- **Over-annotation flag:** sepiapterin reductase activity (GO:0004757) and tetrahydrobiopterin biosynthetic process (GO:0006729) on Q8RJB2 are **IEA:TreeGrafter** inferences from PANTHER PTHR44085, with no experimental support and no known BH4 pathway in *B. cereus*.

Bottom line for the curator: The AlphaFold model is high-confidence and the pocket **does** appear to discriminate sterically in a way consistent with the real kinetics (tight flat aromatics, weak bulky benzil). This supports treating benzil as non-physiological and favouring a small/flat aromatic carbonyl substrate. It does **not**, by itself, prove the native substrate is a quinone — a quinone is one plausible member of the favoured class, alongside other flat aromatic carbonyls and (untested) pterins. The most defensible annotation remains a **general NADPH-dependent aromatic carbonyl reductase of unknown physiological substrate**.

Tools & Database Versions Actually Invoked (provenance)

Tool / resource	Version / date	What was run	Real output (key values)
AlphaFold DB REST API + model	AF-Q8RJB2-F1, model v6, created 2025-08-01	Downloaded model + confidence	global pLDDT 96.8 ; 98% residues very-high. (Seed's <code>..._v4.cif</code> URL now 404s; v6 is current.)
Manual PDB parse (numpy/scipy)	—	Motif ID, pLDDT, pocket map, radial cavity probe	SDR tetrad Asn86/Ser140/Tyr154/ Lys158 (pLDDT 98.0–98.9); Rossmann TGTSQGLG res 7–14 ; pocket 33 res mean pLDDT 97.1; enclosed (7% of 400 rays open), median reach 5.4 Å
PubChem PUG REST 3D	2026-10	Candidate sterics	benzil 10.0 Å, plane-RMSD 0.39 (twisted); 1,4-NQ 5.3 Å, 0.00 (flat); methylglyoxal 3.7 Å; menadione 6.6 Å
Foldseek web server (search.foldseek.com)	pdb100, afdb- swissprot, afdb50 (2026-10)	3Di search	Nearest named: B. subtilis YueD O32099 (40%), yeast IRC24 P40580, then sepiapterin reductases ; PDB: human SPR 1z6z/1nas/6i6v/6i6f/ 7dsf , KRED1 6yc8, yeast YIR035C 6uhx, actinorhodin KR 1x7g, 3-OH- butyrate DH 5b4t/2q2v. No quinone reductase.
UniProtKB REST (<code>Q8RJB2.json</code>)	reviewed; evidence at protein level	EC, GO, family, kinetics	EC 1.1.1.320; PANTHER PTHR44085; Km/kcat table below (ECO:0000269 <code>P 11796169</code>)
kcat/Km computation (pandas)	—	Specificity constants	see kinetics table; 1,4-NQ $3.2\times >$ benzil; best = 1-phenyl-1,2- propanedione

Honest limitations: The AlphaFold model is apo (no NADP/substrate), so pocket metrics are geometric, not energetic; no docking/MD was run. Km/kcat were taken from UniProt's curated ECO:0000269 annotation of

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(full text not retrieved); methylglyoxal and sepiapterin were not among the tested substrates. Foldseek hits are remote (21–28% id), reflecting the common SDR fold.

Kinetics

(**P** 11796169 via UniProt Q8RJB2; NADPH, 37 °C, pH 6.5)

Substrate	Km (μM)	kcat (min^{-1})	kcat/Km ($\text{min}^{-1}\mu\text{M}^{-1}$)	rel. eff. vs benzil	class
1-phenyl-1,2-propanedione	42.0	165	3.93	46.9×	synthetic aryl-alkyl diketone
1,4-naphthoquinone	27.6	7.4	0.268	3.2×	flat fused quinone
benzil	768	64.4	0.084	1.0×	bulky twisted diaryl diketone
1-(4-Me-phenyl)-2-phenyl-diketone	611	37.9	0.062	0.74×	diaryl diketone
1-(4-F-phenyl)-2-phenyl-diketone	584	31.9	0.055	0.65×	diaryl diketone
methyl benzoylformate	1400	16.7	0.012	0.14×	aryl α -ketoester
p-nitrobenzaldehyde	261	1.2	0.005	0.05×	aryl aldehyde

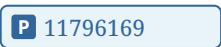
Interpretation: **Lowest K_m → 1,4-naphthoquinone (27.6 μM), tightest binder**; benzil is the weakest binder (768 μM). This is exactly the ordering the seed cited and is **consistent with the enclosed aromatic pocket favouring flat compact aromatics over bulky twisted benzil**. Highest catalytic efficiency, however, is a non-natural aryl-alkyl diketone, so "best substrate" ≠ "quinone." (methylglyoxal, sepiapterin not tested.)

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence limitation
P 11745140	Direct assay / gene isolation	Qualifies	Is benzil physiological?	Gene isolated by screening for benzil→(S)- benzoin	<i>B. cereus</i> → <i>E. coli</i>	High for "benzil = bait"
P 11796169 (Km data)	Direct kinetics	Supports	Km(1,4-NQ) << Km(benzil)?	Km 27.6 vs 768 μM (28× tighter for NQ); kcat/ Km(NQ) 3.2× > benzil	recombinant enzyme in vitro	High; curated ECO:00002
P 11796169 (efficiency)	Direct kinetics	Qualifies	Is a quinone the best substrate?	Best kcat/Km = 1-phenyl-1,2- propanedione (synthetic), not a quinone	in vitro	High; natural substrates untested
P 11796169 (GFP)	Localization	Supports CC	Location	Bipolar cytoplasm	<i>B. cereus</i>	Moderate- high (tagg
P 11796169 (homology)	Evolutionary	Qualifies	Nearest homologs	28–30% to mammalian sepiapterin reductases	cross-species	High; rem (~30%)
P 23640889	Mechanistic (human SPR)	Qualifies	How are quinones turned over?	Quinone redox cycling at cofactor site, not substrate pocket (D257H)	human SPR	High mechanism different enzyme
AlphaFold AF-Q8RJB2- F1 v6	Computational	Supports	Pocket discriminates?	Enclosed aromatic pocket disfavours bulky benzil; consistent with Km order	predicted apo model	High confidence apo, geometric only

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence limitation
PubChem 3D	Computational	Supports	Candidate sterics	benzil bulky/ twisted; 1,4-NQ flat/compact	small- molecule geometry	Geometric no energy
Foldseek	Structural/ evolutionary	Competes	Quinone vs pterin reductase closer?	Sepiapterin reductases & SDR ketoreductases closest; no quinone reductase	PDB100/ AFDB	High; rem hits (gener SDR fold)
UniProt Q8RJB2	Review/ database	Flag	GO annotation quality	GO:0004757 & GO:0006729 are IEA:TreeGrafter only	PANTHER PTHR44085	Database- level

GO Curation Implications (leads — require curator verification)

GO term	Aspect	Current evidence	Recommendation
GO:0004090 carbonyl reductase (NADPH) activity	MF	IEA:RHEA (EC 1.1.1.320)	Retain — best-supported, appropriately general core MF
GO:0016616 oxidoreductase, CH-OH donor, NAD(P) acceptor	MF	EXP	Retain (parent)
GO:0005737 cytoplasm	CC	EXP (GFP, )	Retain
GO:0004757 sepiapterin reductase (NADP+) activity	MF	IEA:TreeGrafter only	Flag / downgrade — no sepiapterin assay for YueD; ~25% id to true SPRs; not core
GO:0006729 tetrahydrobiopterin biosynthetic process	BP	IEA:TreeGrafter only	Flag / consider removal — no BH4 pathway known in <i>B. cereus</i>
(seed-proposed) quinone reductase activity	MF	Km/kcat for 1,4-NQ measured	Do not assert as core. Supported as <i>an</i> in-vitro activity, but not shown to be the physiological role; 1,4-NQ is not the most efficient substrate and structural neighbours are not quinone reductases. If annotated, use with "ISS/IDA for in-vitro 1,4-naphthoquinone reduction" caveat, not as the primary function

Net curation lead: Keep a **general NADPH carbonyl/aryl-ketone reductase MF + cytoplasm CC**; treat **sepiapterin-reductase MF and BH4-biosynthesis BP as non-core phylogenetic over-annotations**; annotate the physiological substrate/process as **unknown**. The structural+kinetic evidence supports "benzil is non-physiological" and "pocket favours flat/compact aromatic carbonyls," but does not justify naming a quinone as the core function.

Mechanistic Scope

Direct activity: **NADPH-dependent reduction of an aromatic carbonyl** (aryl α -diketone/aldehyde/quinone) to the alcohol, by a cytoplasmic SDR using the Ser140–Tyr154–Lys158 relay with Asn86 and an NADP(H) Rossmann site. All in-vitro substrates assayed are synthetic/xenobiotic; "quinone reductase," "sepiapterin reductase," and "tetrahydrobiopterin biosynthesis" are inferences, not demonstrated physiological roles. The pocket's steric preference for flat compact aromatics (low K_m for 1,4-NQ) is a property of the enzyme; which natural flat aromatic carbonyl it acts on in the cell is unknown.

Conflicts and Alternatives

1. **Efficiency vs affinity split:** 1,4-NQ has the lowest K_m but modest k_{cat} ; the most *efficient* substrate (1-phenyl-1,2-propanedione) is a non-natural diketone — so tight binding of a quinone does not by itself make a quinone the physiological substrate.
2. **Family evidence cuts toward pterin/keto-reductase:** PANTHER PTHR44085 (sepiapterin reductase) and Foldseek neighbours argue for pterin/carbonyl chemistry, not a dedicated quinone reductase.
3. **Mechanistic category caveat:** homologous SPR handles quinones by cofactor-site redox cycling
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YueD's defined K_m/k_{cat} suggest true pocket turnover but this was not dissected.
4. **Bacterial quinone size mismatch:** the natural bacterial quinone is **menaquinone** (large lipophilic isoprenoid), which would not fit this small soluble pocket — so "quinone" in vivo is unlikely to be a naphthoquinone-shaped molecule.
5. **Apo-model caveat:** no NADP/substrate in the model; pocket metrics are geometric approximations.
6. **Database carry-over:** sepiapterin/BH4 terms likely propagated across the PANTHER clade (also seen on orthologs).

Knowledge Gaps

Gap	What was checked	Why it matters	Resolution
True physiological substrate	Lit + UniProt kinetics; all tested substrates synthetic	Determines correct MF/BP	Δ yueD metabolomics; screen natural flat aromatic carbonyls (o-quinones, isatin, aromatic aldehydes, pterins)
Pterin activity	Not assayed for YueD	Decides GO:0004757	Direct sepiapterin→BH2 assay with recombinant YueD + NADPH
BH4 pathway in <i>B. cereus</i>	Not established	Decides GO:0006729	Genomic check for GTPCH-I/PTPS; BH4 detection
Quinone mechanism (turnover vs redox cycling)	Km/kcat known; mechanism not dissected	Affects whether "quinone reductase" is a real pocket activity	Steady-state vs single-turnover; O ₂ -consumption/ROS assay with 1,4-NQ
In-cell role	GFP localization only	Separates catalysis from biology	Δ yueD phenotyping under electrophile/oxidative stress

Discriminating Tests (most efficient first)

1. **Δ yueD metabolomics / stress phenotyping** — best route to the native substrate and BP.
2. **Direct sepiapterin and methylglyoxal assays** with recombinant YueD + NADPH — close the two untested candidate classes and settle GO:0004757.
3. **Quinone mechanism assay** (ROS/O₂ consumption vs clean 2-electron reduction for 1,4-NQ) — tests whether the "quinone" activity is pocket turnover or redox cycling.
4. **NADP⁺-bound model + docking** — enables a real steric/electronic discrimination test rather than apo geometry.
5. **Operon/genomic-context analysis** in *B. cereus* — may reveal the served pathway.

Curation Leads (require curator verification)

- **Reference snippets to verify**

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(kinetics, via UniProt ECO:0000269): $K_m = 768 \mu\text{M}$ (benzil) vs $27.6 \mu\text{M}$ (1,4-naphthoquinone); k_{cat}/K_m best for 1-phenyl-1,2-propanedione → confirms NQ binds tighter than benzil but is not the most efficient substrate.

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"Green fluorescent protein-tagged *B. cereus* benzil reductase distributed in the bipolar cytoplasm in *B. cereus* cells." → GO:0005737.

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"Site-directed mutagenesis of the SPR C-terminal substrate-binding site (D257H) completely inhibited sepiapterin reduction but had minimal effects on redox cycling." → quinone activity need not be substrate-pocket-based.

- **Candidate GO actions:** retain GO:0004090, GO:0016616 (MF), GO:0005737 (CC); **downgrade/flag** GO:0004757 (MF) and GO:0006729 (BP) as IEA:TreeGrafter over-annotations; **do not elevate** any quinone-reductase term to core function (in-vitro only).
- **Action change vs seed:** accept "benzil non-physiological" and "pocket discriminates / favours flat aromatic carbonyls"; keep "quinone" as a *plausible but unproven* member of the favoured class, not the established native substrate.
- **Suggested curator question:** Does any source place a natural flat aromatic carbonyl (e.g., an o-quinone, isatin, or pterin) as a *B. cereus* metabolite that YueD could act on in vivo?
- **Suggested experiment:** Δ yueD vs WT metabolomics + targeted activity screen incl. sepiapterin and methylglyoxal.

Supported vs Refuted (summary)

- **Supported:** Benzil is non-physiological (weakest binder, $K_m 768 \mu\text{M}$, synthetic). YueD is a high-confidence cytoplasmic NADPH SDR carbonyl reductase. **$K_m(1,4\text{-NQ}) \ll K_m(\text{benzil})$** , and the enclosed aromatic pocket is **consistent** with that ordering — the pocket does discriminate sterically.

- **Partially supported:** "Native substrate is a naturally occurring aromatic carbonyl" — the enzyme clearly prefers flat/compact aromatic carbonyls, but no natural substrate has been identified.
- **Unresolved / not uniquely supported:** "Most plausibly a quinone" — a quinone fits the favoured class and binds tightly, but the most efficient substrate is a non-natural diketone, structural neighbours are pterin/keto-reductases, and the natural bacterial quinone (menaquinone) is too large. "Pocket discriminates between classes" — supported in the sense that bulky benzil binds poorly and flat aromatics bind well, but an apo model cannot by itself name the physiological substrate.

Limitations

Apo AlphaFold model; geometric (not energetic) sterics; kinetics taken from UniProt's curated annotation (full text not retrieved); methylglyoxal/sepiapterin untested; remote Foldseek hits; single characterization study underlies most wet-lab claims.

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